

MODULE 1.1: INTRODUCTION

Contents

1. Basic Concepts	1
2. Characteristics of Cyber Security.....	2
3. Purposes of Cyber Security	2
4. Importance / Motivation Behind Cyber Security	3
5. Enterprise / Organisation Framework & Functional Areas of Cyber Security	3
6. Components of Cyber Security in Enterprise / Organisation Infrastructure	4
a. 1. Policy.....	4
b. 2. Programmatic.....	4
c. 3. IT Life Cycle.....	5
d. 4. Assessment	6
e. Summary Flow.....	6
7. Five Cybersecurity Challenges.....	6
8. Challenges for Cyber Security Frameworks	7

Basic Concepts

1 Cyber

- Refers to technology that involves data/information, systems, and networks.
- Related to information technology, internet/network, and virtual environments (computers and connected resources).

2 Security

- Means protection against damage, unauthorized access, or vulnerabilities.
- Involves applying technical measures based on the value of data.

3 Enterprise

- Can be an individual, organization, business, or government body.
- Cyber security aims to achieve partial but effective protection with an optimal budget.
- Not necessarily about implementing all best practices or frameworks — focuses on practical balance.

4 Information Security in Context

- Security revolves around data and its value.
- Involves regular monitoring of systems and networks.
- Defense strategies must be designed to mitigate threats.
- Testing of defense strength should be conducted periodically.

Characteristics of Cyber Security

1. **Data Protection Techniques** – Uses encryption, authentication, and access control to safeguard data during creation, storage, and transmission.
2. **Comprehensive Defence** – Focuses on protecting individuals, organizations, and national infrastructure from cyber threats.
3. **Threat Mitigation** – Defends against attacks such as malware, phishing, DDoS, and insider threats.
4. **Unauthorized Access Prevention** – Ensures only legitimate users can access critical systems or data.
5. **CIA Triad Maintenance** – Preserves Confidentiality (no unauthorized access), Integrity (no unauthorized alteration), and Availability (resources always accessible).
6. **Continuous Monitoring** – Involves real-time tracking of systems to detect and respond to anomalies.
7. **Incident Detection and Response** – Includes processes to quickly identify, contain, and recover from security breaches.
8. **Risk Assessment** – Regularly evaluates vulnerabilities and potential impact to prioritize defences.
9. **Policy and Governance** – Follows security frameworks and compliance standards to maintain discipline in protection mechanisms.
10. **Adaptability** – Updates defence systems continuously as threats evolve with new technologies.

Purposes of Cyber Security

1. **Vulnerability Management** – Identifies, analyzes, and reduces weaknesses in software, hardware, and human behaviour.
2. **Threat Management** – Detects and neutralizes malicious activities like hacking, ransomware, or virus infections.
3. **Incident Response Capability** – Prepares for, manages, and recovers from cyber incidents efficiently.
4. **Security Policy Enforcement** – Implements and enforces standards, procedures, baselines, and guidelines across the organization.
5. **Regulatory Compliance** – Ensures adherence to national and international laws, data protection acts, and cyber ethics.
6. **Data Assurance** – Guarantees the reliability, confidentiality, and integrity of data across all operations.
7. **User Awareness and Training** – Educates users about safe online behaviour and phishing prevention.
8. **Business Continuity** – Maintains critical operations even during cyber attacks or system failures.
9. **Incident Documentation** – Records breaches and responses for future prevention and legal compliance.
10. **Collaboration and Coordination** – Promotes cooperation among organizations, law enforcement, and governments against cybercrime.

Importance / Motivation Behind Cyber Security

1. **Financial Protection** – Prevents direct monetary losses and market share decline from cyber attacks.
2. **Data and Identity Safety** – Shields personal and organizational data from theft or misuse.
3. **Reputation Management** – Avoids negative publicity, customer mistrust, and brand damage.
4. **Operational Continuity** – Prevents downtime or disruption in business processes.
5. **Legal Protection** – Helps avoid fines, penalties, or lawsuits arising from data breaches.
6. **Client Retention** – Builds trust and ensures customer data confidentiality.
7. **Prevention of Data Manipulation** – Stops alteration, deletion, or distortion of critical information.
8. **Intellectual Property Protection** – Guards proprietary information, trade secrets, and innovations.
9. **National Security** – Protects government and defence data from cyber espionage or attacks.
10. **Technological Stability** – Maintains the integrity and reliability of modern digital infrastructures.

Enterprise / Organisation Framework & Functional Areas of Cyber Security

Framework Overview

1. A **cyber security framework** helps enterprises defend against attacks.
2. It is **divided into multiple functional areas**, each addressing a specific aspect of security.

Functional Areas

1. **System Administration Security** – Protects operating systems, configurations, and administrative controls.
2. **Network Security** – Safeguards communication channels from unauthorized access or attacks.
3. **Application Security** – Ensures software applications are free from vulnerabilities and exploits.
4. **Endpoint, Server, and Device Security** – Protects individual devices, servers, and connected hardware.
5. **Identity, Authentication, and Access Management** – Manages user identities and access rights securely.
6. **Data Protection & Cryptography** – Secures sensitive data using encryption and key management.
7. **Monitoring, Vulnerability Detection & Patch Management** – Tracks systems, identifies weaknesses, and applies fixes.
8. **High Availability & Disaster Recovery (Physical Protection)** – Ensures continuous service and recovery after failures or disasters.
9. **Incident Response** – Provides structured actions to detect, contain, and recover from security incidents.
10. **Asset Management** – Maintains an updated inventory of IT assets to protect them effectively.
11. **Policy, Audit, and Training** – Establishes rules, conducts audits, and trains staff for security awareness.

Components of Cyber Security in Enterprise / Organisation Infrastructure

Cyber security in an organisation is built around a structured program. The program is divided into four major components:

1. **Policy**
2. **Programmatics**
3. **IT Life Cycle**
4. **Assessment**

1. Policy

A policy is the foundation of any cyber security program. It defines rules, standards, procedures, and guidelines. **Example:** Security for endpoint devices vs. routers will differ, since risks are different.

It answers:

- What needs to be protected?
- To what degree should it be protected?
- What are the consequences of violation?

Enterprise policies must cover:

- Staffing (who is responsible)
- Budget (funding allocation)
- Technology (tools/solutions)

2.

Programmatics

Refers to the supporting elements required for running a cyber security program. It includes People, Budget, and Technology.

2.1 People: People are the core resource of cyber security.

- **Requires management clarity:** clear lines of authority, guidelines, and training (e.g., installing new apps safely).
- Everyone in the organisation must actively participate.
- Grouped based on:
 - Authority (decision makers)
 - Responsibility (who executes tasks)
 - Skill set (technical/non-technical expertise)
 - Limitations (scope and boundaries of work)
- Must work as a team across departments:
 - IT
 - Operations
 - Cyber Security
 - Compliance (what must be done as per rules)
 - Internal Audit

2.2 Budget: Cyber security needs financial planning. Each must be effective and justified. Each budget requirement must be documented

- **Activities:**
 1. Allocate resources and estimate costs (deployment, operation, maintenance).
 2. Define how much money is dedicated to each cyber security item.
 3. Document budget requirements per functional area.

- **Example:**
 - Functional Area 1: Endpoint Security
 - Functional Area 2: Network Security

2.3 Technology

- Motivation: Manual efforts are insufficient because of:
 - Size of enterprise systems
 - Complexity of infrastructure
 - Speed of threats
- Technology focus areas:
 - Choosing the right tools
 - Deciding how to deploy them
 - Planning how to maintain them
- Technology must cover:
 - **Prevention** – block threats before they occur.
 - **Detection** – identify incidents immediately after they happen.
 - **Logging** – record all events locally and at network perimeter.
 - **Response** – follow incident response protocols.
 - **Audit** – maintain transaction-level audits for accountability.

3. IT Life Cycle

Covers the entire span of IT and cyber security integration.

3.1 Strategy

- Defines the plan of action.
- Ensures hardware and software technologies are well-coordinated.
- Cyber security modules must work seamlessly with the IT framework.
- Failure to align strategy leads to loss, inconvenience, and insecure infrastructure.

3.2 Engineering

- Checks if requirements are fully met.
- Ensures systems are installed, configured, and supported properly.
- Products must be suitable for both:
 - Fit for purpose (meets intended goals).
 - Fit for use (usable in real operations).
- Engineering ensures long-term support and scalability.

3.3 Operations

- Cyber security must support daily operations.
- If not properly maintained, security measures become ineffective.
- Involves continuous updates, monitoring, and patching.

4. Assessment

Assessment checks whether the cyber security program actually works.

- **Key Questions:**
 1. What are the current cyber security capabilities?
 2. What risk mitigation techniques are in place?
 3. What operational processes are being followed?
- **Assessment Includes:**
 - Reporting status in case of violations of:
 - Legal requirements (laws/regulations).
 - Compliance requirements (internal/external policies).
 - Regulatory requirements (industry/government standards).
- **Tools and Measures:**
 - Penetration testing
 - Vulnerability assessments
 - Compliance audits
 - Security scorecards

Summary Flow

1. **Policy** – Defines *what, how much, and consequences*.
2. **Programmatics** – People, Budget, Technology.
3. **IT Life Cycle** – Strategy → Engineering → Operations.
4. **Assessment** – Test and verify effectiveness, ensure compliance.

Five Cybersecurity Challenges

Modern cybersecurity infrastructure faces several issues due to advancing technologies, increasing attack sophistication, and complex operational environments. As per NIST SP 800-53 (US standard), the main five challenges are:

1. Scalability

- As organizations grow, the number of systems, users, and devices increases rapidly.
- Managing security consistently across a large environment becomes difficult.
- Each new device or network layer adds potential vulnerabilities.
- Scalable solutions must ensure performance without compromising protection.

2. Judgement Calls

- Security decisions often rely on human judgement — such as assessing risk levels or granting access.
- Different individuals may interpret the same situation differently, leading to inconsistency.
- Overreliance on subjective decisions can create loopholes in policy enforcement.
- Standardized procedures and automation help reduce this issue.

3. Detective Controls

- Detective controls identify incidents *after* they occur (e.g., intrusion detection, monitoring logs).
- Challenge: attackers use stealth techniques that evade detection tools.
- Requires continuous improvement of monitoring tools and real-time analytics.
- False positives or missed alerts can delay response.

4. Security Operations

- Managing daily security activities—patching, monitoring, responding—is complex.
- Integrating multiple tools and coordinating teams is difficult.
- Lack of skilled personnel or clear processes weakens defence.
- Efficient Security Operations Centers (SOCs) and automation tools are essential.

5. Unfulfillable Requirements

- Some security demands are impossible or impractical to meet fully.
- Reasons include limited budget, legacy systems, or conflicting performance goals.
- Example: achieving 100% security without reducing usability is unrealistic.
- Organizations must prioritize feasible and high-impact controls.

Challenges for Cyber Security Frameworks

A cybersecurity framework is effective only when all its components work in proper coordination. Each component faces its own challenges that can affect overall security performance.

1. Policy Framework Challenges

- Difficulty in defining clear, practical, and enforceable security policies.
- Rapidly changing technology makes existing policies outdated.
- Ensuring employee awareness and compliance with policies.
- Balancing security strictness with usability and business needs.

2. Programmatic Framework Challenges

- Shortage of skilled cybersecurity professionals.
- Limited or misallocated budget for security operations.
- Integration issues among people, processes, and technologies.
- Coordination gaps between departments like IT, compliance, and operations.

3. IT Life Cycle Challenges

- Misalignment between IT strategy and cybersecurity planning.
- Poor configuration or outdated engineering practices.
- Operational maintenance not kept up to date, reducing effectiveness.
- Difficulty maintaining long-term support for deployed security products.

4. Assessment Challenges

- Inadequate tools or methods to evaluate real-world security effectiveness.
- Incomplete or inaccurate reporting of violations and risks.
- Failure to meet legal, compliance, or regulatory standards.
- Difficulty in quantifying cybersecurity performance and improvement.